Full Network Devices Configuration

========== HQ-CORE ==========

hostname HQ-CORE

line consol 0

password **CO-CON@admin32026**

login

enable password **CO-Enpr@admin32026**

service password-encryption

username NetAdmin secret **N3C@Admin!2026**

username admin password **123**

ip domain-name we.com

crypto key generate rsa

spanning-tree mode pvst

interface GigabitEthernet1/0/1

no switchport

ip address 105.100.50.2 255.255.255.252

```
interface GigabitEthernet1/0/2
no switchport
ip address 10.20.20.34 255.255.255.252

interface GigabitEthernet1/0/3
no switchport
ip address 10.20.20.38 255.255.255.252

interface GigabitEthernet1/0/4
no switchport
ip address 10.10.10.1 255.255.255.0
ip access-group BLOCK_SERVER out

interface GigabitEthernet1/0/5
shutdown

interface GigabitEthernet1/0/6
shutdown

interface GigabitEthernet1/0/7
shutdown

interface GigabitEthernet1/0/8
shutdown
interface GigabitEthernet1/0/9
```

```
shutdown

interface GigabitEthernet1/0/10
shutdown

interface GigabitEthernet1/0/11
shutdown

interface GigabitEthernet1/0/12
shutdown

interface GigabitEthernet1/0/13
shutdown

interface GigabitEthernet1/0/14
shutdown

interface GigabitEthernet1/0/15
shutdown

interface GigabitEthernet1/0/16
shutdown

interface GigabitEthernet1/0/17
shutdown
```

```
interface GigabitEthernet1/0/18
shutdown

interface GigabitEthernet1/0/19
shutdown

interface GigabitEthernet1/0/20
shutdown

interface GigabitEthernet1/0/21
shutdown

interface GigabitEthernet1/0/22
shutdown

interface GigabitEthernet1/0/23
shutdown

interface GigabitEthernet1/0/24
shutdown

interface GigabitEthernet1/1/1
shutdown

interface GigabitEthernet1/1/2
shutdown
```

```
interface GigabitEthernet1/1/3
shutdown

interface GigabitEthernet1/1/4
shutdown

router ospf 1
log-adjacency-changes
network 10.20.20.32 0.0.0.3 area 0
network 105.100.50.0 0.0.0.3 area 0
network 10.10.10.0 0.0.0.255 area 0
network 10.20.20.36 0.0.0.3 area 0

ip access-list extended BLOCK_SERVER
deny icmp any 10.10.10.0 0.0.0.255
deny tcp any 10.10.10.0 0.0.0.255 eq www
permit ip any any

logging 10.10.10.7
logging trap debugging
service timestamps log datetime msec
ntp server 10.10.10.8

line vty 0 4
exec-timeout 2 0
```

login local

transport input ssh

#copy running-config tftp:

10.10.10.9

HQ-CORE

========== HQ-DSW1==========

hostname HQ-DSW1

line consol 0

password **ML-CON@admin22026**

login

enable password **ML-Enpr@admin22026**

service password-encryption

username NetAdmin secret **N3C@Admin!2026**

username admin password **123**

ip routing

ip arp inspection vlan 10,20,30,40

ip dhcp snooping

ip dhcp snooping vlan 10,20,30,40

```
no ip dhcp snooping information option

ip domain-name we.com
crypto key generate rsa

spanning-tree mode rapid-pvst
spanning-tree vlan 10,20,30,40 priority 4096

interface Port-channel1
switchport mode trunk

interface GigabitEthernet1/0/1
no switchport
ip address 10.20.20.33 255.255.255.252

interface GigabitEthernet1/0/2
ip dhcp snooping trust
switchport mode trunk
ip arp inspection trust
channel-group 1 mode passive

interface GigabitEthernet1/0/3
ip dhcp snooping trust
switchport mode trunk
ip arp inspection trust
channel-group 1 mode passive
```

```
interface GigabitEthernet1/0/4
ip dhcp snooping trust
switchport mode trunk
ip arp inspection trust
channel-group 1 mode passive

interface GigabitEthernet1/0/5
shutdown

interface GigabitEthernet1/0/6
shutdown

interface GigabitEthernet1/0/7
switchport mode trunk

interface GigabitEthernet1/0/8
switchport mode trunk

interface GigabitEthernet1/0/9
switchport mode trunk

interface GigabitEthernet1/0/10
switchport mode trunk

interface GigabitEthernet1/0/11
```

```
shutdown

interface GigabitEthernet1/0/12
shutdown

interface GigabitEthernet1/0/13
shutdown

interface GigabitEthernet1/0/14
shutdown

interface GigabitEthernet1/0/15
shutdown

interface GigabitEthernet1/0/16
shutdown

interface GigabitEthernet1/0/17
shutdown

interface GigabitEthernet1/0/18
shutdown

interface GigabitEthernet1/0/19
shutdown
```

```
interface GigabitEthernet1/0/20
shutdown
interface GigabitEthernet1/0/21
shutdown

interface GigabitEthernet1/0/22
shutdown

interface GigabitEthernet1/0/23
shutdown

interface GigabitEthernet1/0/24
shutdown

interface GigabitEthernet1/1/1
shutdown

interface GigabitEthernet1/1/2
shutdown

interface GigabitEthernet1/1/3
shutdown

interface GigabitEthernet1/1/4
shutdown
```

```
interface Vlan10
ip address 192.168.10.1 255.255.255.0
ip helper-address 10.10.10.6
standby 10 ip 192.168.10.10
standby 10 priority 110
standby 10 preempt

interface Vlan20
ip address 192.168.20.1 255.255.255.0
ip helper-address 10.10.10.6
standby 20 ip 192.168.20.10
standby 20 priority 110
standby 20 preempt

interface Vlan30
ip address 192.168.30.1 255.255.255.0
ip helper-address 10.10.10.6
standby 30 ip 192.168.30.10
standby 30 priority 110
standby 30 preempt

interface Vlan40
ip address 192.168.40.1 255.255.255.0
ip helper-address 10.10.10.6
standby 30 ip 192.168.30.10
standby 40 ip 192.168.40.10
```

```
standby 40 priority 110
standby 40 preempt

router ospf 1
network 10.20.20.32 0.0.0.3 area 0
network 192.168.10.0 0.0.0.255 area 0
network 192.168.20.0 0.0.0.255 area 0
network 192.168.30.0 0.0.0.255 area 0
network 192.168.40.0 0.0.0.255 area 0

logging 10.10.10.7
logging trap debugging
service timestamps log datetime msec
ntp server 10.10.10.8


line vty 0 4
exec-timeout 2 0
login local
transport input ssh

#copy running-config tftp:
10.10.10.9
HQ-DSW1
```

========== HQ-DSW2==========

hostname HQ-DSW2

line consol 0

password **ML-CON@admin22026**

login

enable password **ML-Enpr@admin22026**

service password-encryption

username NetAdmin secret **N3C@Admin!2026**

username admin password **123**

ip routing

ip dhcp snooping vlan 10,20,30,40

no ip dhcp snooping information option

ip dhcp snooping

ip domain-name we.com

crypto key generate rsa

spanning-tree mode rapid-pvst

```
spanning-tree vlan 10,20,30,40 priority 8192

interface Port-channel1
switchport mode trunk

interface GigabitEthernet1/0/1
no switchport
ip address 10.20.20.37 255.255.255.252

interface GigabitEthernet1/0/2
switchport mode trunk
channel-group 1 mode active

interface GigabitEthernet1/0/3
switchport mode trunk
channel-group 1 mode active

interface GigabitEthernet1/0/4
switchport mode trunk
channel-group 1 mode active

interface GigabitEthernet1/0/5
shutdown

interface GigabitEthernet1/0/6
shutdown
```

```
interface GigabitEthernet1/0/7
switchport mode trunk

interface GigabitEthernet1/0/8
switchport mode trunk

interface GigabitEthernet1/0/9
switchport mode trunk

interface GigabitEthernet1/0/10
switchport mode trunk
shutdown

interface GigabitEthernet1/0/11
shutdown

interface GigabitEthernet1/0/12
shutdown

interface GigabitEthernet1/0/13
shutdown

interface GigabitEthernet1/0/14
shutdown
```

```
interface GigabitEthernet1/0/15
shutdown

interface GigabitEthernet1/0/16
shutdown

interface GigabitEthernet1/0/17
shutdown

interface GigabitEthernet1/0/18
shutdown

interface GigabitEthernet1/0/19
shutdown

interface GigabitEthernet1/0/20
shutdown

interface GigabitEthernet1/0/21
shutdown

interface GigabitEthernet1/0/22
shutdown

interface GigabitEthernet1/0/23
shutdown
```

```
interface GigabitEthernet1/0/24
shutdown
interface GigabitEthernet1/1/1
shutdown

interface GigabitEthernet1/1/2
shutdown

interface GigabitEthernet1/1/3
shutdown

interface GigabitEthernet1/1/4
shutdown

interface Vlan10
ip address 192.168.10.2 255.255.255.0
ip helper-address 10.10.10.6
standby 10 ip 192.168.10.10

interface Vlan20
ip address 192.168.20.2 255.255.255.0
ip helper-address 10.10.10.6
standby 20 ip 192.168.20.10

interface Vlan30
```

```
ip address 192.168.30.2 255.255.255.0
ip helper-address 10.10.10.6
standby 30 ip 192.168.30.10
interface Vlan40
ip address 192.168.40.2 255.255.255.0
ip helper-address 10.10.10.6
standby 40 ip 192.168.40.10

router ospf 1
network 10.20.20.36 0.0.0.3 area 0
network 192.168.10.0 0.0.0.255 area 0
network 192.168.20.0 0.0.0.255 area 0
network 192.168.30.0 0.0.0.255 area 0
network 192.168.40.0 0.0.0.255 area 0

line vty 0 4
exec-timeout 2 0
login local
transport input ssh

logging 10.10.10.7
logging trap debugging
service timestamps log datetime msec
ntp server 10.10.10.8

#copy running-config tftp:
```

10.10.10.9

HQ-DSW2

========== HQ-ASW1==========

hostname HQ-ASW1

line consol 0

password **AC-CON@admin12026**

login

enable password **AC-Enpr@admin12026**

service password-encryption

username NetAdmin secret **N3C@Admin!2026**

username admin password **123**

ip domain-name we.com

crypto key generate rsa

ip arp inspection vlan 10

ip dhcp snooping vlan 10

no ip dhcp snooping information option

ip dhcp snooping

```
spanning-tree mode rapid-pvst
spanning-tree extend system-id

interface FastEthernet0/1
switchport access vlan 10
switchport mode access
switchport port-security
switchport port-security mac-address sticky
switchport port-security mac-address sticky 0060.5C1D.52D5
spanning-tree portfast

interface FastEthernet0/2
switchport access vlan 10
switchport mode access
switchport port-security
switchport port-security mac-address sticky
switchport port-security mac-address sticky 00E0.8F16.D179
spanning-tree portfast

interface FastEthernet0/3
switchport access vlan 10
switchport mode access
switchport port-security
switchport port-security mac-address sticky
switchport port-security mac-address sticky 0060.5C6C.E909
```

```
spanning-tree portfast

interface FastEthernet0/4
switchport access vlan 10
switchport mode access
switchport port-security
switchport port-security mac-address sticky
switchport port-security mac-address sticky 0002.160E.DCB0
spanning-tree portfast

interface FastEthernet0/5
ip arp inspection trust
ip dhcp snooping trust
switchport mode trunk

interface FastEthernet0/6
ip arp inspection trust
ip dhcp snooping trust
switchport mode trunk

interface FastEthernet0/7
shutdown

interface FastEthernet0/8
shutdown
```

```
interface FastEthernet0/9
shutdown

interface FastEthernet0/10
shutdown

interface FastEthernet0/11
shutdown

interface FastEthernet0/12
shutdown

interface FastEthernet0/13
shutdown

interface FastEthernet0/14
shutdown

interface FastEthernet0/15
shutdown

interface FastEthernet0/16
shutdown

interface FastEthernet0/17
shutdown
```

```
interface FastEthernet0/18
shutdown

interface FastEthernet0/19
shutdown

interface FastEthernet0/20
shutdown

interface FastEthernet0/21
shutdown

interface FastEthernet0/22
shutdown

interface FastEthernet0/23
shutdown

interface FastEthernet0/24
shutdown

interface GigabitEthernet0/1
shutdown

interface GigabitEthernet0/2
```

```
shutdown

interface Vlan1
no ip address
shutdown

interface Vlan10
ip address 192.168.10.3 255.255.255.0

ip default-gateway 192.168.10.10

line vty 0 4
exec-timeout 2 0
login local
transport input ssh

logging 10.10.10.7
logging trap debugging
service timestamps log datetime msec
ntp server 10.10.10.8

#copy running-config tftp:
10.10.10.9
HQ-DSW2
```

========== HQ-ASW2==========

hostname HQ-ASW2

line consol 0

password **AC-CON@admin12026**

login

enable password **AC-Enpr@admin12026**

service password-encryption

username NetAdmin secret **N3C@Admin!2026**

username admin password **123**

ip domain-name we.com

crypto key generate rsa

ip arp inspection vlan 20

ip dhcp snooping vlan 20

no ip dhcp snooping information option

ip dhcp snooping

spanning-tree mode rapid-pvst

spanning-tree extend system-id

interface FastEthernet0/1

```
switchport access vlan 20
switchport mode access
switchport port-security
switchport port-security mac-address sticky
switchport port-security mac-address sticky 000A.4123.2E21
spanning-tree portfast

interface FastEthernet0/2
switchport access vlan 20
switchport mode access
switchport port-security
switchport port-security mac-address sticky
switchport port-security mac-address sticky 00D0.97C7.8169
spanning-tree portfast

interface FastEthernet0/3
switchport access vlan 20
switchport mode access
switchport port-security
switchport port-security mac-address sticky
switchport port-security mac-address sticky 00D0.D35C.53E4
spanning-tree portfast

interface FastEthernet0/4
switchport access vlan 20
switchport mode access
```

```
switchport port-security
switchport port-security mac-address sticky
switchport port-security mac-address sticky 0001.42B9.BAEE
spanning-tree portfast

interface FastEthernet0/5
ip arp inspection trust
ip dhcp snooping trust
switchport mode trunk

interface FastEthernet0/6
ip arp inspection trust
ip dhcp snooping trust
switchport mode trunk

interface FastEthernet0/7
shutdown

interface FastEthernet0/8
shutdown

interface FastEthernet0/9
shutdown
interface FastEthernet0/10
shutdown
```

```
interface FastEthernet0/11
shutdown

interface FastEthernet0/12
shutdown

interface FastEthernet0/13
shutdown

interface FastEthernet0/14
shutdown

interface FastEthernet0/15
shutdown

interface FastEthernet0/16
shutdown

interface FastEthernet0/17
shutdown

interface FastEthernet0/18
shutdown

interface FastEthernet0/19
shutdown
```

```
interface FastEthernet0/20
shutdown

interface FastEthernet0/21
shutdown

interface FastEthernet0/22
shutdown

interface FastEthernet0/23
shutdown

interface FastEthernet0/24
shutdown

interface GigabitEthernet0/1
shutdown

interface GigabitEthernet0/2
shutdown

interface Vlan20
ip address 192.168.20.3 255.255.255.0

ip default-gateway 192.168.20.10
```

line vty 0 4

exec-timeout 2 0

login local

transport input ssh

logging 10.10.10.7

logging trap debugging

service timestamps log datetime msec

ntp server 10.10.10.8

#copy running-config tftp:

10.10.10.9

HQ-DSW2

========== HQ-ASW3==========

hostname HQ-ASW3

line consol 0

password **AC-CON@admin12026**

login

enable password **AC-Enpr@admin12026**

service password-encryption

```
username NetAdmin secret N3C@Admin!2026
username admin password 123

ip domain-name we.com
crypto key generate rsa


ip arp inspection vlan 30

ip dhcp snooping vlan 30
no ip dhcp snooping information option
ip dhcp snooping

spanning-tree mode rapid-pvst
spanning-tree extend system-id

interface FastEthernet0/1
switchport access vlan 30
switchport mode access
switchport port-security
switchport port-security mac-address sticky
switchport port-security mac-address sticky 0001.6447.4B5D
spanning-tree portfast

interface FastEthernet0/2
switchport access vlan 30
```

```
switchport mode access
switchport port-security
switchport port-security mac-address sticky
switchport port-security mac-address sticky 0001.6311.6486
spanning-tree portfast

interface FastEthernet0/3
switchport access vlan 30
switchport mode access
switchport port-security
switchport port-security mac-address sticky
switchport port-security mac-address sticky 0001.9626.072E
spanning-tree portfast

interface FastEthernet0/4
switchport access vlan 30
switchport mode access
switchport port-security
switchport port-security mac-address sticky
switchport port-security mac-address sticky 0060.3E17.E59D
spanning-tree portfast

interface FastEthernet0/5
ip arp inspection trust
ip dhcp snooping trust
switchport mode trunk
```

```
interface FastEthernet0/6
ip arp inspection trust
ip dhcp snooping trust
switchport mode trunk

interface FastEthernet0/7
shutdown

interface FastEthernet0/8
shutdown

interface FastEthernet0/9
shutdown

interface FastEthernet0/10
shutdown

interface FastEthernet0/11
shutdown

interface FastEthernet0/12
shutdown

interface FastEthernet0/13
shutdown
```

```
interface FastEthernet0/14
shutdown

interface FastEthernet0/15
shutdown

interface FastEthernet0/16
shutdown

interface FastEthernet0/17
shutdown

interface FastEthernet0/18
shutdown

interface FastEthernet0/19
shutdown

interface FastEthernet0/20
shutdown

interface FastEthernet0/21
shutdown

interface FastEthernet0/22
```

```
shutdown

interface FastEthernet0/23
shutdown

interface FastEthernet0/24
shutdown

interface GigabitEthernet0/1
shutdown

interface GigabitEthernet0/2
shutdown

interface Vlan30
ip address 192.168.30.3 255.255.255.0

ip default-gateway 192.168.30.10

line vty 0 4
exec-timeout 2 0
login local
transport input ssh

logging 10.10.10.7
logging trap debugging
```

service timestamps log datetime msec

ntp server 10.10.10.8

#copy running-config tftp:

10.10.10.9

HQ-DSW3

========== HQ-ASW4==========

hostname HQ-ASW4

line consol 0

password **AC-CON@admin12026**

login

enable password **AC-Enpr@admin12026**

service password-encryption

username NetAdmin secret **N3C@Admin!2026**

username admin password **123**

ip domain-name we.com

crypto key generate rsa

ip arp inspection vlan 40

```
ip dhcp snooping vlan 40
no ip dhcp snooping information option
ip dhcp snooping

spanning-tree mode rapid-pvst
spanning-tree extend system-id

interface FastEthernet0/1
switchport access vlan 40
switchport mode access
switchport port-security
switchport port-security mac-address sticky
switchport port-security mac-address sticky 0007.ECB1.35B0
spanning-tree portfast

interface FastEthernet0/2
switchport access vlan 40
switchport mode access
switchport port-security
switchport port-security mac-address sticky
switchport port-security mac-address sticky 00E0.8F89.6503
spanning-tree portfast

interface FastEthernet0/3
switchport access vlan 40
switchport mode access
```

```
switchport port-security
switchport port-security mac-address sticky
switchport port-security mac-address sticky 00E0.8F08.D257
spanning-tree portfast

interface FastEthernet0/4
switchport access vlan 40
switchport mode access
switchport port-security
switchport port-security mac-address sticky
switchport port-security mac-address sticky 0001.649A.6317
spanning-tree portfast

interface FastEthernet0/5
ip arp inspection trust
ip dhcp snooping trust
switchport mode trunk

interface FastEthernet0/6
ip arp inspection trust
ip dhcp snooping trust
switchport mode trunk

interface FastEthernet0/7
shutdown
```

```
interface FastEthernet0/8
shutdown

interface FastEthernet0/9
shutdown

interface FastEthernet0/10
shutdown

interface FastEthernet0/11
shutdown

interface FastEthernet0/12
shutdown

interface FastEthernet0/13
shutdown

interface FastEthernet0/14
shutdown

interface FastEthernet0/15
shutdown

interface FastEthernet0/16
shutdown
```

```
interface FastEthernet0/17
shutdown

interface FastEthernet0/18
shutdown

interface FastEthernet0/19
shutdown

interface FastEthernet0/20
shutdown

interface FastEthernet0/21
shutdown

interface FastEthernet0/22
shutdown

interface FastEthernet0/23
shutdown

interface FastEthernet0/24
shutdown

interface GigabitEthernet0/1
shutdown
```

```
interface GigabitEthernet0/2
shutdown

interface Vlan40
ip address 192.168.40.3 255.255.255.0

ip default-gateway 192.168.40.10

line vty 0 4
exec-timeout 2 0
login local
transport input ssh

logging 10.10.10.7
logging trap debugging
service timestamps log datetime msec
ntp server 10.10.10.8

#copy running-config tftp:
10.10.10.9
HQ-DSW3
```

========== HQ-ASW5==========

hostname HQ-ASW5

line consol 0

password **AC-CON@admin12026**

login

enable password **AC-Enpr@admin12026**

service password-encryption

username NetAdmin secret **N3C@Admin!2026**

username admin password **123**

ip domain-name we.com

crypto key generate rsa

ip arp inspection vlan 10,20,30,40

ip dhcp snooping vlan 10,20,30,40

no ip dhcp snooping information option

ip dhcp snooping

spanning-tree mode rapid-pvst

spanning-tree extend system-id

```
interface FastEthernet0/1
ip dhcp snooping trust
switchport mode access
switchport port-security
switchport port-security mac-address sticky
switchport port-security mac-address sticky 0000.0C2E.0209
spanning-tree portfast

interface FastEthernet0/2
switchport mode access
switchport port-security
switchport port-security mac-address sticky
switchport port-security mac-address sticky 000D.BD72.639D
spanning-tree portfast

interface FastEthernet0/3
switchport mode access
switchport port-security
switchport port-security mac-address sticky
switchport port-security mac-address sticky 00E0.F9E2.357B
spanning-tree portfast

interface FastEthernet0/4
shutdown
interface FastEthernet0/5
switchport mode access
```

```
switchport port-security
switchport port-security mac-address sticky
switchport port-security mac-address sticky 0030.A3A0.40B2
spanning-tree portfast

interface FastEthernet0/6
ip arp inspection trust

interface FastEthernet0/7
shutdown

interface FastEthernet0/8
shutdown

interface FastEthernet0/9
shutdown

interface FastEthernet0/10
shutdown

interface FastEthernet0/11
shutdown

interface FastEthernet0/12
shutdown
```

```
interface FastEthernet0/13
shutdown

interface FastEthernet0/14
shutdown

interface FastEthernet0/15
shutdown

interface FastEthernet0/16
shutdown

interface FastEthernet0/17
shutdown

interface FastEthernet0/18
shutdown

interface FastEthernet0/19
shutdown

interface FastEthernet0/20
shutdown

interface FastEthernet0/21
shutdown
```

```
interface FastEthernet0/22
shutdown

interface FastEthernet0/23
shutdown

interface FastEthernet0/24
shutdown

interface GigabitEthernet0/1
shutdown

interface GigabitEthernet0/2
shutdown

interface Vlan1
ip address 10.10.10.2 255.255.255.0


line vty 0 4
exec-timeout 2 0
login local
transport input ssh

logging 10.10.10.7
logging trap debugging
```

service timestamps log datetime msec

ntp server 10.10.10.8

#copy running-config tftp:

10.10.10.9

HQ-DSW4

========== R1==========

hostname R1

line consol 0

password **RO-CON@admin42026**

login

enable password **RO-Enpr@admin42026**

service password-encryption

username NetAdmin secret **N3C@Admin!2026**

username admin password **123**

ip domain-name we.com

crypto key generate rsa

spanning-tree mode pvst

```
interface GigabitEthernet0/0
ip address 20.20.20.2 255.255.255.252

interface GigabitEthernet0/1
ip address 105.100.50.1 255.255.255.252

router ospf 1
log-adjacency-changes
network 20.20.20.0 0.0.0.3 area 0
network 105.100.50.0 0.0.0.3 area 0

line vty 0 4
exec-timeout 2 0
login local
transport input ssh

logging 10.10.10.7
logging trap debugging
service timestamps log datetime msec
ntp server 10.10.10.8

#copy running-config tftp:
10.10.10.9
R1
```

========== R2==========

hostname R2

line consol 0

password **RO-CON@admin42026**

login

enable password **RO-Enpr@admin42026**

service password-encryption

username NetAdmin secret **N3C@Admin!2026**

username admin password **123**

ip domain-name we.com

crypto key generate rsa

spanning-tree mode pvst

interface GigabitEthernet0/0

ip address 30.30.30.1 255.255.255.252

interface GigabitEthernet0/1

ip address 20.20.20.1 255.255.255.252

```
router ospf 1
network 30.30.30.0 0.0.0.3 area 0
network 20.20.20.0 0.0.0.3 area 0

line vty 0 4
exec-timeout 2 0
login local
transport input ssh

logging 10.10.10.7
logging trap debugging
service timestamps log datetime msec
ntp server 10.10.10.8

#copy running-config tftp:
10.10.10.9
R2
```

========== R3==========

```
hostname R3
line consol 0
```

password **RO-CON@admin42026**

```
login

enable password
```

RO-Enpr@admin42026

```
service password-encryption

username NetAdmin secret N3C@Admin!2026
username admin password 123

ip domain-name we.com
crypto key generate rsa

spanning-tree mode pvst

interface GigabitEthernet0/0
ip address 205.200.100.1 255.255.255.252

interface GigabitEthernet0/1
ip address 30.30.30.2 255.255.255.252

router ospf 1
network 205.200.100.0 0.0.0.3 area 0
network 30.30.30.0 0.0.0.3 area 0
network 0.0.0.0 255.255.255.255 area 0

line vty 0 4
exec-timeout 2 0
login local
transport input ssh
```

logging 10.10.10.7

logging trap debugging

service timestamps log datetime msec

ntp server 10.10.10.8

#copy running-config tftp:

10.10.10.9

R3

========== BR-CORE==========

hostname BR-CORE

line consol 0

password **CO-CON@admin32026**

login

enable password **CO-Enpr@admin32026**

service password-encryption

username NetAdmin secret **N3C@Admin!2026**

username admin password **123**

ip domain-name we.com

crypto key generate rsa

```
spanning-tree mode pvst

interface GigabitEthernet1/0/1
no switchport
ip address 10.20.20.42 255.255.255.252

interface GigabitEthernet1/0/2
no switchport
ip address 10.20.20.46 255.255.255.252

interface GigabitEthernet1/0/3
no switchport
ip address 205.200.100.2 255.255.255.252

interface GigabitEthernet1/0/4
shutdown

interface GigabitEthernet1/0/5
shutdown

interface GigabitEthernet1/0/6
shutdown

interface GigabitEthernet1/0/7
shutdown
```

```
interface GigabitEthernet1/0/8
shutdown

interface GigabitEthernet1/0/9
shutdown

interface GigabitEthernet1/0/10
shutdown

interface GigabitEthernet1/0/11
shutdown

interface GigabitEthernet1/0/12
shutdown

interface GigabitEthernet1/0/13
shutdown

interface GigabitEthernet1/0/14
shutdown

interface GigabitEthernet1/0/15
shutdown

interface GigabitEthernet1/0/16
shutdown
```

```
interface GigabitEthernet1/0/17
shutdown

interface GigabitEthernet1/0/18
shutdown

interface GigabitEthernet1/0/19
shutdown

interface GigabitEthernet1/0/20
shutdown

interface GigabitEthernet1/0/21
shutdown

interface GigabitEthernet1/0/22
shutdown

interface GigabitEthernet1/0/23
shutdown

interface GigabitEthernet1/0/24
shutdown

interface GigabitEthernet1/1/1
shutdown
```

```
interface GigabitEthernet1/1/2
shutdown

interface GigabitEthernet1/1/3
shutdown

interface GigabitEthernet1/1/4
shutdown

router ospf 1
network 10.20.20.40 0.0.0.3 area 0
network 10.20.20.44 0.0.0.3 area 0
network 205.200.100.0 0.0.0.3 area 0
network 0.0.0.0 255.255.255.255 area 0

line vty 0 4
exec-timeout 2 0
login local
transport input ssh

logging 10.10.10.7
logging trap debugging
service timestamps log datetime msec
ntp server 10.10.10.8

#copy running-config tftp:
```

10.10.10.9

BR-CORE

========== BR-DSW1==========

hostname BR-DSW1

line consol 0

password **ML-CON@admin22026**

login

enable password **ML-Enpr@admin22026**

service password-encryption

username NetAdmin secret **N3C@Admin!2026**

username admin password **123**

ip dhcp excluded-address 192.168.60.1 192.168.60.50

ip dhcp excluded-address 192.168.70.1 192.168.70.50

ip dhcp excluded-address 192.168.80.1 192.168.80.50

!

ip dhcp pool VLAN60

network 192.168.60.0 255.255.255.0

default-router 192.168.60.10

```
dns-server 8.8.8.8

ip dhcp pool VLAN70
network 192.168.70.0 255.255.255.0
default-router 192.168.70.10
dns-server 8.8.8.8

ip dhcp pool VLAN80
network 192.168.80.0 255.255.255.0
default-router 192.168.80.10
dns-server 8.8.8.8

ip routing

spanning-tree mode rapid-pvst
spanning-tree vlan 10,20,30,40 priority 4096

interface Port-channel1
switchport mode trunk

interface GigabitEthernet1/0/1
switchport mode trunk
channel-group 1 mode desirable

interface GigabitEthernet1/0/2
switchport mode trunk
```

```
channel-group 1 mode desirable

interface GigabitEthernet1/0/3
switchport mode trunk

interface GigabitEthernet1/0/4
switchport mode trunk

interface GigabitEthernet1/0/5
switchport mode trunk

interface GigabitEthernet1/0/6
no switchport
ip address 10.20.20.45 255.255.255.252

interface GigabitEthernet1/0/7
shutdown

interface GigabitEthernet1/0/8
shutdown

interface GigabitEthernet1/0/9
shutdown

interface GigabitEthernet1/0/10
shutdown
```

```
interface GigabitEthernet1/0/11
shutdown

interface GigabitEthernet1/0/12
shutdown

interface GigabitEthernet1/0/13
shutdown

interface GigabitEthernet1/0/14
shutdown

interface GigabitEthernet1/0/15
shutdown

interface GigabitEthernet1/0/16
shutdown

interface GigabitEthernet1/0/17
shutdown

interface GigabitEthernet1/0/18
shutdown

interface GigabitEthernet1/0/19
shutdown
```

```
interface GigabitEthernet1/0/20
shutdown

interface GigabitEthernet1/0/21
shutdown

interface GigabitEthernet1/0/22
shutdown

interface GigabitEthernet1/0/23
shutdown

interface GigabitEthernet1/0/24
shutdown

interface GigabitEthernet1/1/1
shutdown

interface GigabitEthernet1/1/2
shutdown

interface GigabitEthernet1/1/3
shutdown

interface GigabitEthernet1/1/4
shutdown
```

```
interface Vlan60
mac-address 0060.47cb.3a01
ip address 192.168.60.1 255.255.255.0
standby 60 ip 192.168.60.10
standby 60 priority 110
standby 60 preempt

interface Vlan70
mac-address 0060.47cb.3a02
ip address 192.168.70.1 255.255.255.0
standby 70 ip 192.168.70.10
standby 70 priority 110
standby 70 preempt

interface Vlan80
mac-address 0060.47cb.3a03
ip address 192.168.80.1 255.255.255.0
standby 80 ip 192.168.80.10
standby 80 priority 110
standby 80 preempt

router ospf 1
network 192.168.60.0 0.0.0.255 area 0
network 192.168.70.0 0.0.0.255 area 0
network 192.168.80.0 0.0.0.255 area 0
network 10.20.20.44 0.0.0.3 area 0
```

line vty 0 4

exec-timeout 2 0

login local

transport input ssh

logging 10.10.10.7

logging trap debugging

service timestamps log datetime msec

ntp server 10.10.10.8

#copy running-config tftp:

10.10.10.9

BR-DSW1

========== BR-DSW2==========

hostname BR-DWS2

line consol 0

password **ML-CON@admin22026**

login

enable password **ML-Enpr@admin22026**

service password-encryption

username NetAdmin secret **N3C@Admin!2026**

```
username admin password 123


ip dhcp excluded-address 192.168.60.1 192.168.60.50
ip dhcp excluded-address 192.168.70.1 192.168.70.50
ip dhcp excluded-address 192.168.80.1 192.168.80.50

ip dhcp pool VLAN60
network 192.168.60.0 255.255.255.0
default-router 192.168.60.10
dns-server 8.8.8.8

ip dhcp pool VLAN70
network 192.168.70.0 255.255.255.0
default-router 192.168.70.10
dns-server 8.8.8.8

ip dhcp pool VLAN80
network 192.168.80.0 255.255.255.0
default-router 192.168.80.10
dns-server 8.8.8.8

ip routing

spanning-tree mode rapid-pvst
spanning-tree vlan 10,20,30,40 priority 8192
```

```
interface Port-channel1
switchport mode trunk

interface GigabitEthernet1/0/1
switchport mode trunk
channel-group 1 mode auto

interface GigabitEthernet1/0/2
switchport mode trunk
channel-group 1 mode auto

interface GigabitEthernet1/0/3
switchport mode trunk

interface GigabitEthernet1/0/4
switchport mode trunk

interface GigabitEthernet1/0/5
switchport mode trunk

interface GigabitEthernet1/0/6
no switchport
ip address 10.20.20.41 255.255.255.252

interface GigabitEthernet1/0/7
shutdown
```

```
interface GigabitEthernet1/0/8
shutdown

interface GigabitEthernet1/0/9
shutdown

interface GigabitEthernet1/0/10
shutdown

interface GigabitEthernet1/0/11
shutdown

interface GigabitEthernet1/0/12
shutdown

interface GigabitEthernet1/0/13
shutdown

interface GigabitEthernet1/0/14
shutdown

interface GigabitEthernet1/0/15
shutdown

interface GigabitEthernet1/0/16
shutdown
```

```
interface GigabitEthernet1/0/17
shutdown

interface GigabitEthernet1/0/18
shutdown

interface GigabitEthernet1/0/19
shutdown

interface GigabitEthernet1/0/20
shutdown

interface GigabitEthernet1/0/21
shutdown

interface GigabitEthernet1/0/22
shutdown

interface GigabitEthernet1/0/23
shutdown

interface GigabitEthernet1/0/24
shutdown

interface GigabitEthernet1/1/1
shutdown
```

```
interface GigabitEthernet1/1/2
shutdown

interface GigabitEthernet1/1/3
shutdown

interface GigabitEthernet1/1/4
shutdown

interface Vlan60
mac-address 0004.9a30.a801
ip address 192.168.60.1 255.255.255.0
standby 60 ip 192.168.60.10

interface Vlan70
mac-address 0004.9a30.a802
ip address 192.168.70.2 255.255.255.0
standby 70 ip 192.168.70.10

interface Vlan80
mac-address 0004.9a30.a803
ip address 192.168.80.2 255.255.255.0
standby 80 ip 192.168.80.10

router ospf 1
log-adjacency-changes
```

```
network 192.168.60.0 0.0.0.255 area 0
network 192.168.70.0 0.0.0.255 area 0
network 192.168.80.0 0.0.0.255 area 0
network 10.20.20.40 0.0.0.3 area 0

line vty 0 4
exec-timeout 2 0
login local
transport input ssh

logging 10.10.10.7
logging trap debugging
service timestamps log datetime msec
ntp server 10.10.10.8

#copy running-config tftp:
10.10.10.9
BR-DSW2
```

========== BR-ASW1==========

hostname BR-ASW1

line consol 0

password **AC-CON@admin12026**

login

enable password **AC-Enpr@admin12026**

service password-encryption

username NetAdmin secret **N3C@Admin!2026**

username admin password **123**

ip domain-name we.com

crypto key generate rsa

ip arp inspection vlan 60

ip dhcp snooping vlan 60

no ip dhcp snooping information option

ip dhcp snooping

spanning-tree mode rapid-pvst

spanning-tree extend system-id

```
interface FastEthernet0/1
switchport access vlan 60
switchport mode access
switchport port-security
switchport port-security mac-address sticky
switchport port-security violation restrict
switchport port-security mac-address sticky 0005.5E48.D861
spanning-tree portfast

interface FastEthernet0/2
switchport access vlan 60
switchport mode access
switchport port-security
switchport port-security mac-address sticky
switchport port-security violation restrict
switchport port-security mac-address sticky 0001.43DB.0480
spanning-tree portfast

interface FastEthernet0/3
switchport access vlan 60
switchport mode access
switchport port-security
switchport port-security mac-address sticky
switchport port-security violation restrict
switchport port-security mac-address sticky 0030.A302.E593
spanning-tree portfast
```

```
interface FastEthernet0/4
switchport access vlan 60
switchport mode access
switchport port-security
switchport port-security mac-address sticky
switchport port-security violation restrict
switchport port-security mac-address sticky 0005.5E16.7C10
spanning-tree portfast

interface FastEthernet0/5
ip arp inspection trust
ip dhcp snooping trust
switchport mode trunk

interface FastEthernet0/6
ip arp inspection trust
ip dhcp snooping trust
switchport mode trunk

interface FastEthernet0/7
shutdown

interface FastEthernet0/8
shutdown

interface FastEthernet0/9
```

```
shutdown

interface FastEthernet0/10
shutdown

interface FastEthernet0/11
shutdown

interface FastEthernet0/12
shutdown

interface FastEthernet0/13
shutdown

interface FastEthernet0/14
shutdown

interface FastEthernet0/15
shutdown

interface FastEthernet0/16
shutdown

interface FastEthernet0/17
shutdown
```

```
interface FastEthernet0/18
shutdown

interface FastEthernet0/19
shutdown

interface FastEthernet0/20
shutdown

interface FastEthernet0/21
shutdown

interface FastEthernet0/22
shutdown

interface FastEthernet0/23
shutdown

interface FastEthernet0/24
shutdown

interface GigabitEthernet0/1
shutdown

interface GigabitEthernet0/2
shutdown
```

```
interface Vlan60
ip address 192.168.60.3 255.255.255.0

ip default-gateway 192.168.60.10

line vty 0 4
exec-timeout 2 0
login local
transport input ssh

logging 10.10.10.7
logging trap debugging
service timestamps log datetime msec
ntp server 10.10.10.8

#copy running-config tftp:
10.10.10.9
BR-ASW1
```

========== BR-ASW2==========

hostname BR-ASW2

line consol 0

password **AC-CON@admin12026**

login

enable password **AC-Enpr@admin12026**

service password-encryption

username NetAdmin secret **N3C@Admin!2026**

username admin password **123**

ip domain-name we.com

crypto key generate rsa

ip arp inspection vlan 70

ip dhcp snooping vlan 70

no ip dhcp snooping information option

ip dhcp snooping

spanning-tree mode rapid-pvst

spanning-tree extend system-id

```
interface FastEthernet0/1
switchport access vlan 70
switchport mode access
switchport port-security
switchport port-security mac-address sticky
switchport port-security violation restrict
switchport port-security mac-address sticky 0001.C95A.347A
spanning-tree portfast

interface FastEthernet0/2
switchport access vlan 70
switchport mode access
switchport port-security
switchport port-security mac-address sticky
switchport port-security violation restrict
switchport port-security mac-address sticky 000A.F316.E58C
spanning-tree portfast

interface FastEthernet0/3
switchport access vlan 70
switchport mode access
switchport port-security
switchport port-security mac-address sticky
switchport port-security violation restrict
spanning-tree portfast
```

```
interface FastEthernet0/4
switchport access vlan 70
switchport mode access
switchport port-security
switchport port-security mac-address sticky
switchport port-security violation restrict
switchport port-security mac-address sticky 0001.9602.AAE9
spanning-tree portfast

interface FastEthernet0/5
ip arp inspection trust
ip dhcp snooping trust
switchport mode trunk

interface FastEthernet0/6
ip arp inspection trust
ip dhcp snooping trust
switchport mode trunk

interface FastEthernet0/7
shutdown

interface FastEthernet0/8
shutdown

interface FastEthernet0/9
```

```
shutdown

interface FastEthernet0/10
shutdown

interface FastEthernet0/11
shutdown

interface FastEthernet0/12
shutdown

interface FastEthernet0/13
shutdown

interface FastEthernet0/14
shutdown

interface FastEthernet0/15
shutdown

interface FastEthernet0/16
shutdown

interface FastEthernet0/17
shutdown
```

```
interface FastEthernet0/18
shutdown

interface FastEthernet0/19
shutdown

interface FastEthernet0/20
shutdown

interface FastEthernet0/21
shutdown

interface FastEthernet0/22
shutdown

interface FastEthernet0/23
shutdown

interface FastEthernet0/24
shutdown

interface GigabitEthernet0/1
shutdown

interface GigabitEthernet0/2
shutdown
```

```
interface Vlan70
ip address 192.168.70.3 255.255.255.0

ip default-gateway 192.168.70.10

line vty 0 4
exec-timeout 2 0
login local
transport input ssh

logging 10.10.10.7
logging trap debugging
service timestamps log datetime msec
ntp server 10.10.10.8

#copy running-config tftp:
10.10.10.9
BR-ASW2
```

========== BR-ASW3==========

hostname BR-ASW3

line consol 0

password **AC-CON@admin12026**

login

enable password **AC-Enpr@admin12026**

service password-encryption

username NetAdmin secret **N3C@Admin!2026**

username admin password **123**

ip domain-name we.com

crypto key generate rsa

ip arp inspection vlan 80

ip dhcp snooping vlan 80

no ip dhcp snooping information option

ip dhcp snooping

spanning-tree mode rapid-pvst

spanning-tree extend system-id

```
interface FastEthernet0/1
switchport access vlan 80
switchport mode access
switchport port-security
switchport port-security mac-address sticky
switchport port-security violation restrict
switchport port-security mac-address sticky 000A.41D5.C13A
spanning-tree portfast

interface FastEthernet0/2
switchport access vlan 80
switchport mode access
switchport port-security
switchport port-security mac-address sticky
switchport port-security violation restrict
switchport port-security mac-address sticky 0030.F2CD.107C
spanning-tree portfast

interface FastEthernet0/3
switchport access vlan 80
switchport mode access
switchport port-security
switchport port-security mac-address sticky
switchport port-security violation restrict
spanning-tree portfast
```

```
interface FastEthernet0/4
switchport access vlan 80
switchport mode access
switchport port-security
switchport port-security mac-address sticky
switchport port-security violation restrict
switchport port-security mac-address sticky 0007.EC93.9D5B
spanning-tree portfast

interface FastEthernet0/5
ip arp inspection trust
ip dhcp snooping trust
switchport mode trunk

interface FastEthernet0/6
ip arp inspection trust
ip dhcp snooping trust
switchport mode trunk

interface FastEthernet0/7
shutdown

interface FastEthernet0/8
shutdown

interface FastEthernet0/9
```

```
shutdown

interface FastEthernet0/10
shutdown

interface FastEthernet0/11
shutdown

interface FastEthernet0/12
shutdown

interface FastEthernet0/13
shutdown

interface FastEthernet0/14
shutdown

interface FastEthernet0/15
shutdown

interface FastEthernet0/16
shutdown

interface FastEthernet0/17
shutdown
```

```
interface FastEthernet0/18
shutdown

interface FastEthernet0/19
shutdown

interface FastEthernet0/20
shutdown

interface FastEthernet0/21
shutdown

interface FastEthernet0/22
shutdown

interface FastEthernet0/23
shutdown

interface FastEthernet0/24
shutdown

interface GigabitEthernet0/1
shutdown

interface GigabitEthernet0/2
shutdown
```

```
interface Vlan80
ip address 192.168.80.3 255.255.255.0

ip default-gateway 192.168.80.10

line vty 0 4
exec-timeout 2 0
login local
transport input ssh

logging 10.10.10.7
logging trap debugging
service timestamps log datetime msec
ntp server 10.10.10.8

#copy running-config tftp:
10.10.10.9
BR-ASW3
```